Diffractive Beam Shaper Design via Finite Element Method

Mathematical Description of the Method

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1 The Problem

Design a **diffractive phase element** (DPE) that transforms an input beam f_1 into a desired output intensity pattern $|f_2|^2$. The DPE is described by its transmission function

$$T_{\mathrm{DPE}}(x, y) = \exp\big(i\,\phi(x, y)\big),$$

where $\phi(x, y)$ is the phase function to be computed. Since $|T_{\mathrm{DPE}}| = 1$, no energy is absorbed.

In the optical setup, the input and output planes are at focal distance f from a Fourier-transforming lens. Wave propagation is described by the 2D Fourier transform:

$$f_2 = \mathcal{F}\big\{T_{\mathrm{DPE}} \cdot f_1\big\}.$$

Since only the **intensity** $|f_2|^2$ matters (not its phase), there is a large design freedom that the algorithms below exploit.

2 The Method

2.1 Step 1: Mesh Generation

Lay topologically equivalent meshes over the input plane E and output plane A. For **rectangular topology**, nodes are indexed by (i, j) with $i, j \in \{0, \ldots, s-1\}$, forming quadrilateral cells. Interior nodes have 4 direct neighbors and 8 surrounding nodes.

Interior node positions are initialized using **4-point interpolation** (Atkin, 1994):

$$P_0 = \sum_{l=1}^{4} g_l\, P_l,$$

where the weights depend on the Euclidean distances in index space:

$$g_1 = \frac{d_2\, d_3\, d_4}{(d_1 + d_2)\,(d_1 d_2 + d_3 d_4)}, \qquad g_2 = \frac{d_1\, d_3\, d_4}{(d_1 + d_2)\,(d_1 d_2 + d_3 d_4)},$$

$$g_3 = \frac{d_1\, d_2\, d_4}{(d_3 + d_4)\,(d_3 d_4 + d_1 d_2)}, \qquad g_4 = \frac{d_1\, d_2\, d_3}{(d_3 + d_4)\,(d_3 d_4 + d_1 d_2)},$$

with $d_k = \big[(i_0 - i_k)^2 + (j_0 - j_k)^2\big]^{1/2}$.

For **triangular topology**, nodes are indexed by (i, j) with $j = 0, \ldots, s-1$ and $i = 0, \ldots, j$, using a 6-point interpolation formula with analogous weights.

2.2 Step 2: Mesh Optimization

2.2.1 Equal-Area Optimization (8-Point Algorithm)

For constant-intensity signals, equal area implies equal energy. For an interior node P_0 surrounded by 8 neighbors $P_1, \ldots, P_8$ (numbered counterclockwise from top: P_1=top, P_2=top-right, ..., P_8=top-left), the **ideal position** $C(x_c, y_c)$ that equalizes the 4 surrounding cell areas satisfies:

$$x_c = \frac{bf - de}{bc - ad}, \qquad y_c = \frac{af - ce}{bc - ad},$$

where

$$\begin{aligned}
a &= y_2 - y_8 - y_6 + y_4, & b &= x_2 - x_8 - x_6 + x_4,\\
c &= y_8 - y_6 - y_4 + y_2, & d &= x_8 - x_6 - x_4 + x_2,\\
e &= x_1(y_2 - y_8) + x_5(y_4 - y_6) + y_1(x_8 - x_2) + y_5(x_6 - x_4),\\
f &= x_3(y_2 - y_4) + x_7(y_8 - y_6) + y_3(x_4 - x_2) + y_7(x_6 - x_8).
\end{aligned}$$

The node is moved iteratively: $P_0' = P_0 + \alpha\,(C - P_0)$ with step size $0 < \alpha \leq 1$.

2.2.2 Energy-Weighted Optimization (General Signals)

For signals with non-uniform intensity, the Python implementation uses **energy-weighted Laplace smoothing**. Each interior node is moved toward the energy-weighted centroid of its 4 direct neighbors:

$$C = \frac{w_{\text{left}}\,P_{\text{left}} + w_{\text{right}}\,P_{\text{right}} + w_{\text{up}}\,P_{\text{up}} + w_{\text{down}}\,P_{\text{down}}}{w_{\text{left}} + w_{\text{right}} + w_{\text{up}} + w_{\text{down}}},$$

where w_{left} is the total energy in the two cells to the left of the node, etc. This pulls nodes toward high-energy regions, creating smaller cells there and larger cells in low-energy regions. A validity check rejects any move that would create a negative-area (tangled) cell.

The energy in each quadrilateral cell is computed by scanning pixels inside the polygon using a ray-casting point-in-polygon test.

2.3 Step 3: Phase Recovery (Polynomial Least Squares)

The mesh correspondences $\text{in}(i,j) = (x_{ij}, y_{ij})$ and $\text{out}(i,j) = (\phi_x^{ij}, \phi_y^{ij})$ define a discrete geometric transformation. By the **stationary phase condition**:

$$\nabla\phi(x_{ij}, y_{ij}) = c \cdot \begin{pmatrix} \phi_x^{ij} - \frac{1}{2} \\ \phi_y^{ij} - \frac{1}{2} \end{pmatrix}, \qquad c = 2\pi N,$$

where N is the FFT grid size and the subtraction of $\frac{1}{2}$ centers the output coordinates (matching the `fftshift` convention).

We approximate ϕ as a bivariate polynomial of degree $D-1$:

$$\phi(x, y) = \sum_{\substack{k,l \geq 0 \\ k+l \leq D-1}} a_{kl}\, x^k y^l,$$

with $D(D+1)/2$ unknown coefficients. The gradient conditions give a linear system:

$$M \cdot \mathbf{a} = \mathbf{b},$$

where the matrix entries are

$$M_{\text{row,col}} = \sum_{i,j} \Big[m \cdot k \cdot x_{ij}^{m-1+k-1}\, y_{ij}^{n+l} + n \cdot l \cdot x_{ij}^{m+k}\, y_{ij}^{n-1+l-1} \Big],$$

and the right-hand side entries are

$$b_{\text{row}} = \sum_{i,j} \Big[k \cdot \phi_x^{ij} \cdot x_{ij}^{k-1}\, y_{ij}^{l} + l \cdot \phi_y^{ij} \cdot x_{ij}^{k}\, y_{ij}^{l-1} \Big],$$

using the index mapping $\text{row} = l(2D-l+1)/2 + k$ and $\text{col} = n(2D-n+1)/2 + m$. This system is solved via least squares.

2.4 Step 4: IFTA Refinement

The phase from Step 3 serves as a starting point for the **Iterative Fourier Transform Algorithm** (Gerchberg–Saxton/Fienup), which alternates between input and output planes:

- **Input plane**: replace amplitude with the known $|f_1|$, keep phase.
- **Output plane**: replace amplitude with desired $|f_2|$ (phase freedom), keep phase.

The Studienarbeit runs 10 iterations for efficiency, then 20 for SNR.

2.5 The SepOp Baseline

The Separabilisierungsoperator maps any 2D signal to a nearby separable signal:

$$\text{SepOp}\colon\ f(x,y) \ \mapsto\ \underbrace{\left(\int f(\xi, y)\, d\xi \right)}_{p(y)} \cdot \underbrace{\left(\int f(x, \xi)\, d\xi \right)}_{q(x)}.$$

In the discrete case: $q(x) = \sum_y f(x,y)$ and $p(y) = \sum_x f(x,y)$, giving $s(x,y) = q(x) \cdot p(y)$. For separable signals, the 2D problem decomposes into two independent 1D problems with analytical solutions.

3 Quality Metrics

Signal-to-Noise Ratio (SNR):

$$\text{SNR}_I(f_2, f) = \frac{\|f\|^2}{\||f_2| - \alpha|f|\|^2}, \qquad \alpha = \frac{\langle |f|, |f_2| \rangle}{\|f\|^2}.$$

Diffraction Efficiency:

$$\eta = \frac{\|f\|^2_{W_{\text{Signal}}}}{\|f_1\|^2_{W_{\text{DE}}}}.$$

4 Original Results (1997)

Input → Output	**Method**	**SNR (dB)**	**η (%)**
Circle → Marilyn	SepOp	19.3	78.6
	Kugel	16.9	75.1
	Random	13.8	68.9
	FEM	**20.8**	**79.0**
Insep. → Marilyn	SepOp	32.8	92.1
	FEM	**35.8**	86.0
Gauss → Marilyn	**SepOp**	**31.4**	**92.5**
	FEM	29.9	83.0
Quadrat → Marilyn	SepOp	20.8	79.1
	FEM	20.5	78.0

The FEM method excels for non-separable input signals and performs comparably to SepOp for separable ones.

5 References

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